

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Oracle Database • Oracle SQL • Python/Pandas • Power BI

1,000	\$258,978.84	\$258.98	3.75 / 5
TRANSACTIONS	TOTAL REVENUE	AVERAGE PURCHASE	AVERAGE RATING

1. Executive Summary

This project delivers an end-to-end customer analytics solution using the supplied customer shopping behavior dataset. The analysis is designed around an Oracle Database workflow, with Python used for data validation and preparation, Oracle SQL used for business analysis, and Power BI used for interactive reporting.

The dataset contains **1,000 transaction records** and the analysis produces a total recorded purchase value of **\$258,978.84**, with an average transaction value of **\$258.98** and an average review rating of **3.75**.

2. Business Objectives

- Measure overall revenue, transaction volume and average customer purchase value.
- Identify high-performing categories, products, locations, seasons and payment methods.
- Understand subscription and shipping behavior and compare customer value across groups.
- Segment customers using previous purchase behavior for retention and loyalty strategies.
- Provide Oracle SQL outputs that can be directly consumed by a Power BI dashboard.

3. Technology & Workflow

Stage	Technology	Purpose
1. Source	Excel	Customer shopping transaction dataset
2. Preparation	Python / Pandas	Validation, profiling, derived age groups
3. Database	Oracle Database	Central analytical table and SQL processing
4. Analysis	Oracle SQL	Aggregations, CASE, ranking, business KPIs
5. Visualization	Power BI	Interactive dashboard and management reporting
6. Decision Support	Business Insights	Recommendations based on observed patterns

4. Project Deliverables

Primary deliverable: Oracle-based Customer Shopping Behavior Analytics Report. **Supporting deliverable:** Oracle SQL query script covering KPI, category, product, customer, subscription, shipping, location, age and discount analysis.

5. Dataset Profiling & Data Preparation

Attribute	Details
Records	1,000 transactions
Original columns	18 source attributes
Customer fields	Customer ID, Age, Gender, Location
Product fields	Item Purchased, Category, Size, Color
Transaction fields	Purchase Amount, Season, Review Rating
Behavior fields	Subscription, Shipping, Discount, Promo, Previous Purchases, Payment, Frequency
Derived field	Age Group for analytical reporting

6. Python / Pandas Preparation

- Loaded the Excel source into Pandas and verified the worksheet structure.
- Checked numerical fields including purchase amount, age, previous purchases and review rating.
- Profiled categorical fields such as category, season, subscription status, shipping type and payment method.
- Created an analytical **age_group** field using 18–24, 25–34, 35–44, 45–54 and 55–64 bands.
- Prepared Oracle-friendly column names such as purchase_amount_usd, subscription_status and previous_purchases.
- Produced grouped summaries that can be reproduced in Oracle SQL and visualized in Power BI.

7. Oracle Database Design

The recommended Oracle table is **CUSTOMER_SHOPPING_BEHAVIOR**. Customer ID is treated as the transaction/customer identifier from the supplied dataset. Numeric measures are stored using NUMBER types, while descriptive attributes use VARCHAR2.

Oracle Column	Recommended Type	Business Meaning
CUSTOMER_ID	VARCHAR2(20)	Customer identifier
AGE	NUMBER(3)	Customer age
GENDER	VARCHAR2(20)	Gender category
ITEM_PURCHASED	VARCHAR2(50)	Purchased product
CATEGORY	VARCHAR2(30)	Product category
PURCHASE_AMOUNT_USD	NUMBER(10,2)	Transaction value
LOCATION	VARCHAR2(50)	Customer location
SEASON	VARCHAR2(20)	Purchase season
REVIEW_RATING	NUMBER(3,2)	Customer rating
SUBSCRIPTION_STATUS	VARCHAR2(5)	Subscriber status
SHIPPING_TYPE	VARCHAR2(30)	Shipping method
DISCOUNT_APPLIED	VARCHAR2(5)	Discount indicator
PREVIOUS_PURCHASES	NUMBER(5)	Purchase history
PAYMENT_METHOD	VARCHAR2(30)	Payment type
FREQUENCY_OF_PURCHASES	VARCHAR2(20)	Purchase frequency

8. Oracle SQL Architecture

Raw/Source → **CUSTOMER_SHOPPING_BEHAVIOR** → **Aggregation Queries** → **Business Result Sets** → **Power BI**. This architecture keeps business calculations centralized in Oracle and makes the dashboard easier to refresh and audit.

9. Revenue by Category

Category	Transactions	Revenue	Avg Purchase
Clothing	181	\$50,186.62	\$277.27
Beauty	172	\$44,502.17	\$258.73
Electronics	174	\$43,664.85	\$250.95
Footwear	164	\$43,518.99	\$265.36
Home	155	\$39,721.63	\$256.27
Accessories	154	\$37,384.58	\$242.76

Management view: Clothing is the leading category by recorded revenue at **\$50,186.62**. Category-level revenue should guide inventory planning, promotional placement and dashboard monitoring.

10. Revenue by Gender

Gender	Transactions	Revenue	Avg Purchase
Female	478	\$123,311.08	\$257.97
Male	522	\$135,667.76	\$259.90

11. Subscription Performance

Subscription	Transactions	Revenue	Avg Purchase
No	628	\$165,224.50	\$263.10
Yes	372	\$93,754.34	\$252.03

The non-subscriber group represents **628** transactions versus **372** subscriber transactions. This makes subscription conversion an important business opportunity.

12. Shipping Performance

Shipping Type	Transactions	Revenue	Avg Purchase
Express	218	\$59,393.00	\$272.44
Free Shipping	256	\$66,973.80	\$261.62
Standard	254	\$64,443.92	\$253.72
Next Day Air	272	\$68,168.12	\$250.62

Observation: Express has the highest average purchase among shipping types at **\$272.44**.

13. Discount Impact

Discount Applied	Transactions	Revenue	Avg Purchase
No	592	\$154,799.14	\$261.49
Yes	408	\$104,179.70	\$255.34

14. Customer Segmentation

Segment	Customers	Revenue	Avg Purchase
Loyal	652	\$165,493.81	\$253.82
Returning	322	\$86,045.77	\$267.22
New	26	\$7,439.26	\$286.13

The segmentation rule is transparent and reproducible in Oracle using a CASE expression based on Previous Purchases. This supports targeted retention and loyalty campaigns.

15. Age Group Performance

Age Group	Transactions	Revenue	Avg Purchase
18–24	148	\$38,653.67	\$261.17
25–34	210	\$53,802.38	\$256.20
35–44	237	\$62,112.84	\$262.08
45–54	203	\$50,956.51	\$251.02
55–64	202	\$53,453.44	\$264.62

16. Top 10 Products by Revenue

Product	Orders	Revenue	Avg Rating
Moisturizer	33	\$10,440.65	3.85
Pillow	35	\$9,940.77	3.69
Loafers	37	\$9,930.59	3.74
Smartphone	43	\$9,917.32	3.58
Perfume	38	\$9,888.14	3.81
Sweater	38	\$9,600.19	3.69
Bedsheet	36	\$9,560.78	3.78
Face Cream	40	\$9,526.52	3.68
Boots	34	\$9,346.79	3.97
Smartwatch	30	\$9,090.86	3.65

17. Location Performance — Top 8

Location	Transactions	Revenue	Avg Purchase
Florida	111	\$31,315.76	\$282.12
Ohio	117	\$29,683.39	\$253.70
Massachusetts	112	\$29,063.63	\$259.50
Georgia	102	\$28,801.51	\$282.37
Washington	95	\$24,060.77	\$253.27
New York	94	\$23,843.13	\$253.65
Texas	96	\$23,822.26	\$248.15
Arizona	95	\$23,562.04	\$248.02

18. Seasonal Performance

Season	Transactions	Revenue	Avg Purchase
Summer	274	\$72,912.22	\$266.10
Fall	247	\$62,646.06	\$253.63
Spring	244	\$62,333.63	\$255.47
Winter	235	\$61,086.93	\$259.94

Highest seasonal revenue: Summer with \$72,912.22.

19. Payment Method Performance

Payment Method	Transactions	Revenue	Avg Purchase
Cash	212	\$57,769.32	\$272.50
Debit Card	212	\$54,542.22	\$257.27
Credit Card	194	\$50,300.26	\$259.28
Venmo	199	\$48,870.87	\$245.58
PayPal	183	\$47,496.17	\$259.54

20. Purchase Frequency

Frequency	Transactions	Revenue	Avg Purchase
Weekly	210	\$57,501.92	\$273.82
Annually	211	\$55,339.92	\$262.27
Quarterly	189	\$48,059.67	\$254.28
Monthly	197	\$49,850.00	\$253.05
Bi-Weekly	193	\$48,227.33	\$249.88

Highest average purchase frequency group: Weekly at \$273.82 per transaction.

21. Oracle SQL Business Questions

1. What is the total revenue and average purchase amount?
2. Which category generates the highest revenue?
3. Which products are the top performers by revenue?
4. How do subscribers and non-subscribers compare?
5. Which shipping type has the highest average transaction value?
6. Which locations generate the most revenue?
7. Which age groups contribute the most revenue?
8. What is the relationship between previous purchases and customer segment?
9. Does discount usage change average purchase value?
10. Which products rank in the top three within each category?

22. Power BI Dashboard Specification

Dashboard Area	Visual	Business Purpose
Executive KPIs	4 KPI Cards	Revenue, transactions, average purchase, average rating
Category	Clustered Bar	Compare category revenue and sales volume
Product	Top-N Bar	Identify top products by revenue
Customer	Donut / Bar	Subscription and customer segment distribution
Age	Column / Bar	Revenue contribution by age group
Geography	Map / Bar	Location-level revenue comparison
Season	Column Chart	Seasonal revenue and average purchase
Shipping	Bar Chart	Shipping type and transaction value
Payment	Bar / Donut	Payment method distribution
Filters	Slicers	Category, Gender, Season, Subscription, Location, Shipping

23. Business Insights

- **Revenue scale:** The dataset records **\$258,978.84** in purchase value across 1,000 transactions.
- **Customer value:** The average transaction is **\$258.98**, providing a baseline for customer-value monitoring.
- **Product performance:** Clothing is the leading category by revenue.
- **Subscription:** The non-subscriber population is much larger than the subscriber population, creating a conversion opportunity.
- **Shipping:** Express records the highest average purchase among shipping types.
- **Seasonality:** Summer is the highest-revenue season in the supplied data.
- **Frequency:** Weekly has the highest average transaction value among frequency groups.

24. Business Recommendations

Subscription Conversion: Create targeted offers for non-subscribers and use exclusive benefits to convert high-value customers.

Loyalty Strategy: Reward customers with stronger purchase histories and create tiered loyalty benefits.

Category Investment: Prioritize high-revenue categories for inventory, campaign placement and cross-selling.

Discount Optimization: Evaluate discount programs using revenue and profitability together rather than transaction volume alone.

Location Targeting: Allocate marketing budgets toward higher-revenue locations and test localized campaigns.

Seasonal Planning: Align inventory and promotional calendars with the strongest revenue season.

Shipping Strategy: Evaluate premium shipping offers for higher-value transactions and monitor margin impact.

25. Final Project Conclusion

The project demonstrates a professional Data Analyst workflow using the user's customer shopping dataset: Python/Pandas for preparation, Oracle Database for centralized storage, Oracle SQL for business analysis, and Power BI for decision-focused reporting. The resulting analysis provides a repeatable framework for monitoring revenue, customer behavior, product performance, subscriptions, discounts, shipping and geographic performance.

Portfolio Project Title: Customer Shopping Behavior Analysis using Oracle SQL, Python & Power BI

Core Skills Demonstrated: Excel/Pandas data preparation, Oracle SQL aggregation, GROUP BY, CASE, analytic functions, ROW_NUMBER, ranking, customer segmentation, KPI design, Power BI data modeling and business storytelling.

26. Oracle SQL Query Library

Overall KPI

```
SELECT COUNT(*) total_transactions, SUM(purchase_amount_usd) total_revenue, ROUND(AVG(purchase_amount_usd),2) avg_purchase,
ROUND(AVG(review_rating),2) avg_rating FROM customer_shopping_behavior;
```

Category Revenue

```
SELECT category, COUNT(*) transactions, ROUND(SUM(purchase_amount_usd),2) revenue FROM customer_shopping_behavior GROUP BY category ORDER
BY revenue DESC;
```

Top 10 Products

```
SELECT item_purchased, COUNT(*) orders, ROUND(SUM(purchase_amount_usd),2) revenue FROM customer_shopping_behavior GROUP BY item_purchased
ORDER BY revenue DESC FETCH FIRST 10 ROWS ONLY;
```

Subscription

```
SELECT subscription_status, COUNT(*) transactions, ROUND(SUM(purchase_amount_usd),2) revenue, ROUND(AVG(purchase_amount_usd),2)
avg_purchase FROM customer_shopping_behavior GROUP BY subscription_status;
```

Customer Segmentation

```
SELECT CASE WHEN previous_purchases=0 THEN 'New' WHEN previous_purchases BETWEEN 1 AND 10 THEN 'Returning' ELSE 'Loyal' END segment,
COUNT(*) customers, ROUND(SUM(purchase_amount_usd),2) revenue FROM customer_shopping_behavior GROUP BY CASE WHEN previous_purchases=0 THEN
'New' WHEN previous_purchases BETWEEN 1 AND 10 THEN 'Returning' ELSE 'Loyal' END;
```

Top 3 per Category

```
SELECT * FROM (SELECT category,item_purchased,SUM(purchase_amount_usd) revenue, ROW_NUMBER() OVER(PARTITION BY category ORDER BY
SUM(purchase_amount_usd) DESC) rn FROM customer_shopping_behavior GROUP BY category,item_purchased) WHERE rn<=3;
```

27. Recommended Oracle-to-Power BI Model

Use Oracle as the governed source of truth. Create either a reporting view or a curated table containing the required dimensions and measures. Connect Power BI to Oracle, verify data types, create measures for Total Revenue, Transactions, Average Purchase and Average Rating, then add slicers and drill-through pages.

Measure	DAX / Logic
Total Revenue	SUM(customer_shopping_behavior[purchase_amount_usd])
Transactions	COUNTROWS(customer_shopping_behavior)
Average Purchase	AVERAGE(customer_shopping_behavior[purchase_amount_usd])
Average Rating	AVERAGE(customer_shopping_behavior[review_rating])